$$\left(Cl_2 + H_2O \longrightarrow HCl + HOCl\right) \times 3$$

$$\left(HCl + NaOH \longrightarrow NaCl + H_2O\right) \times 3$$

$$\left(HOCl + NaOH \longrightarrow NaOCl + H_2O\right) \times 3$$

$$3NaOCl \longrightarrow NaClO_3 + 2NaCl$$

overall reaction

$$\boxed{3Cl_2 + 6NaOH \longrightarrow \underset{\text{sodium chlorate}}{NaClO_3} + 5NaCl + 3H_2O}$$

Oxidising and bleaching action: Chlorine is a strong oxidising and bleaching agent because of the nascent oxygen.

$$H_2O + Cl_2 \longrightarrow HCl + \underset{\text{Hypo chlorous acid}}{HOCl}$$

$$HOCl \longrightarrow HCl + (O)$$

Colouring matter + Nascent oxygen → Colourless oxidation product

The bleaching of chlorine is permanent. It oxidises ferrous salts to ferric, sulphites to sulphates and hydrogen sulphide to sulphur.

$$2FeCl_2 + Cl_2 \longrightarrow 2FeCl_3$$

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

$$2FeSO_4 + H_2SO_4 + HOCl \longrightarrow Fe_2(SO_4)_3 + HCl + H_2O$$

overall reaction

$$\boxed{2FeSO_4 + H_2SO_4 + Cl_2 \longrightarrow Fe_2(SO_4)_3 + 2HCl}$$

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

$$Na_2SO_3 + HOCl \longrightarrow Na_2SO_4 + HCl$$

overall reaction

$$\boxed{Na_2SO_3 + H_2O + Cl_2 \longrightarrow Na_2SO_4 + 2HCl}$$

$$Cl_2 + H_2S \longrightarrow 2HCl + S$$

Preparation of bleaching powder: Bleaching powder is produced by passing chlorine gas through dry slaked lime (calcium hydroxide).

$$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$$

Displacement redox reactions: Chlorine displaces bromine from bromides and iodine from iodide salts.

$$Cl_2 + 2KBr \longrightarrow 2KCl + Br_2$$

$$Cl_2 + 2KI \longrightarrow 2KCl + I_2$$

Formation of addition compounds: Chlorine forms addition products with sulphur dioxide, carbon monoixde and ethylene. It forms substituted products with alkanes/arenes.

$$SO_2 + Cl_2 \longrightarrow \underset{\text{Sulphuryl chloride}}{SO_2Cl_2}$$

$$CO + Cl_2 \longrightarrow \underset{\text{Carbonyl chloride}}{COCl_2}$$

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